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The Editor-in-Chief
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Dear Editor,

We submit the manuscript “**Machine Learning Models for Mental-Health Prediction from Actigraphy: A Systematic Appraisal and an Empirical Demonstration**” for consideration as a Research Paper in *Artificial Intelligence in Medicine*.

Why this paper. The open actigraphy cohorts DEPRESJON, PSYKOSE and HYPERAKTIV, now harmonised as OBF-Psychiatric, are the most used public benchmarks for machine-learning detection of mental disorders from wearables, and the accuracies reported on them have climbed from the dataset authors’ leave-one-patient-out 0.7 to above 0.9. Our paper asks whether that is progress. Part I is a closed-scope systematic appraisal of the 28 published models on these cohorts with PROBAST+AI and TRIPOD+AI, using outcomes that can be checked against the data: unit of train/test splitting, external validation, calibration reporting, participant-level events per predictor, code availability, and double counting of the control group that DEPRESJON and PSYKOSE share. Part II re-evaluates the cohorts with models fixed a priori, ten seeds and participant-level bootstrap intervals under the designs the appraisal scores as high and low risk.

Main findings. Record-wise splitting inflates AUROC by up to 0.17 and turns a chance-level ADHD classifier into an apparently useful one; frozen cross-cohort transfer loses up to 0.21 AUROC in one direction with collapse of calibration; calibration is poor even under honest validation, is not repairable from the training cohort, and is never reported in the literature on these cohorts; honest performance is stable across feature subsets, resampling and cross-validation schemes. Every published claim above the honest band coincides with a day-level or unstated splitting unit.

Fit to the journal. The paper is a methodological contribution to the evaluation of AI prediction models in medicine, provides reference baselines for widely used clinical benchmarks, and follows TRIPOD+AI as authors. All analysis code, run manifests and generated tables are public, and the figures were regenerated independently on Google Colab from the repository with identical results. The TRIPOD+AI checklist and feature definitions are supplied as supplementary material.

Declarations. The manuscript is original, is not under consideration elsewhere, and has not been published in any form. Both authors approved the submission. No funding was received. We declare no conflicts of interest. The study is a secondary analysis of publicly available, de-identified datasets released for research use by their custodians. We choose the subscription publication route.

Suggested reviewers. Methodologists from the PROBAST/TRIPOD community and researchers in digital phenotyping of mood and psychotic disorders would be well placed to assess the work; we have no exclusions.

Thank you for considering our manuscript.

Yours sincerely,

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